

ESP32 RFID Anti-Wall-Jumping System

Advanced multi-sensor security solution for perimeter protection

Overview

The **ESP32 RFID Anti-Wall-Jumping System** is a robust, real-time perimeter security prototype designed to detect unauthorized access (like wall climbing or jumping) and respond instantly. It leverages an Active IR Beam sensor for reliable perimeter detection, a long-range UHF RFID reader for authorized access, a DFPlayer Mini for audio alarms, and a SIM800L module for instant SMS notifications to security personnel.

The system features a **built-in Captive Portal (Web UI)** for easy configuration on the fly, avoiding the need to re-flash the firmware to change settings.

Key Features

-  **Active Perimeter Detection:** Uses an Active Single-Beam IR Detector for extreme weather-resistant and long-range intrusion detection (replacing old ultrasonic sensors).
-  **Instant SMS Alerts:** Uses SIM800L to send immediate notifications to a registered emergency contact upon intrusion.
-  **Audio Alarms:** Integrates DFPlayer Mini to broadcast custom localized voice alarms or sirens.
-  **Long Range RFID (UHF WG26):** Allows authorized personnel (using the RD906M Directional Panel Antenna) to bypass the system from a distance.
-  **Local SD Card Logging:** All events and access attempts are saved to an onboard MicroSD card in FAT32 format.
-  **Web-Based Captive Portal:** Built-in async web server allowing administrators to configure system thresholds, modes, and sensors over Wi-Fi without compiling code.
-  **Real-Time Clock:** Built-in DS3231 RTC for accurate time-stamping of intrusion logs.
-  **Hardened Firmware:** Production-grade concurrency protection with dedicated FreeRTOS mutexes, deferred restart handling, validated RTC input, and guarded SD operations.

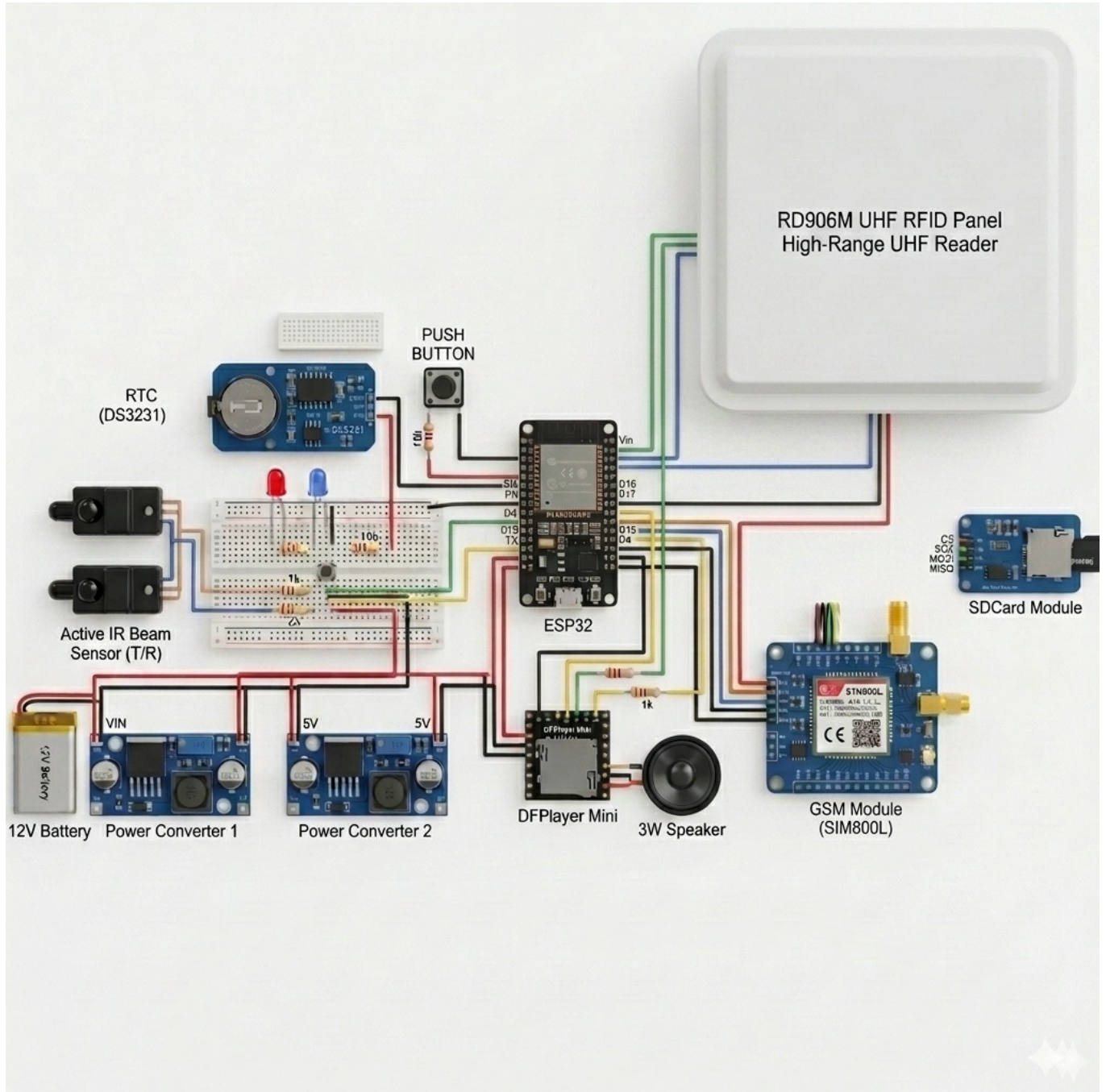


<i>Component</i>	<i>Function</i>
ESP32-38P	<i>Core processing and Web Server AP</i>
RD906M UHF Panel	<i>Long-range RFID Scanning</i>
Active IR Beam	<i>Perimeter breach / motion detection</i>
SIM800L	<i>GSM Module for SMS alerts</i>
DFPlayer Mini	<i>MP3 Audio playback</i>
DS3231	<i>Precision RTC for logging</i>
MicroSD Module	<i>Local event storage</i>

Realistic Component Layout

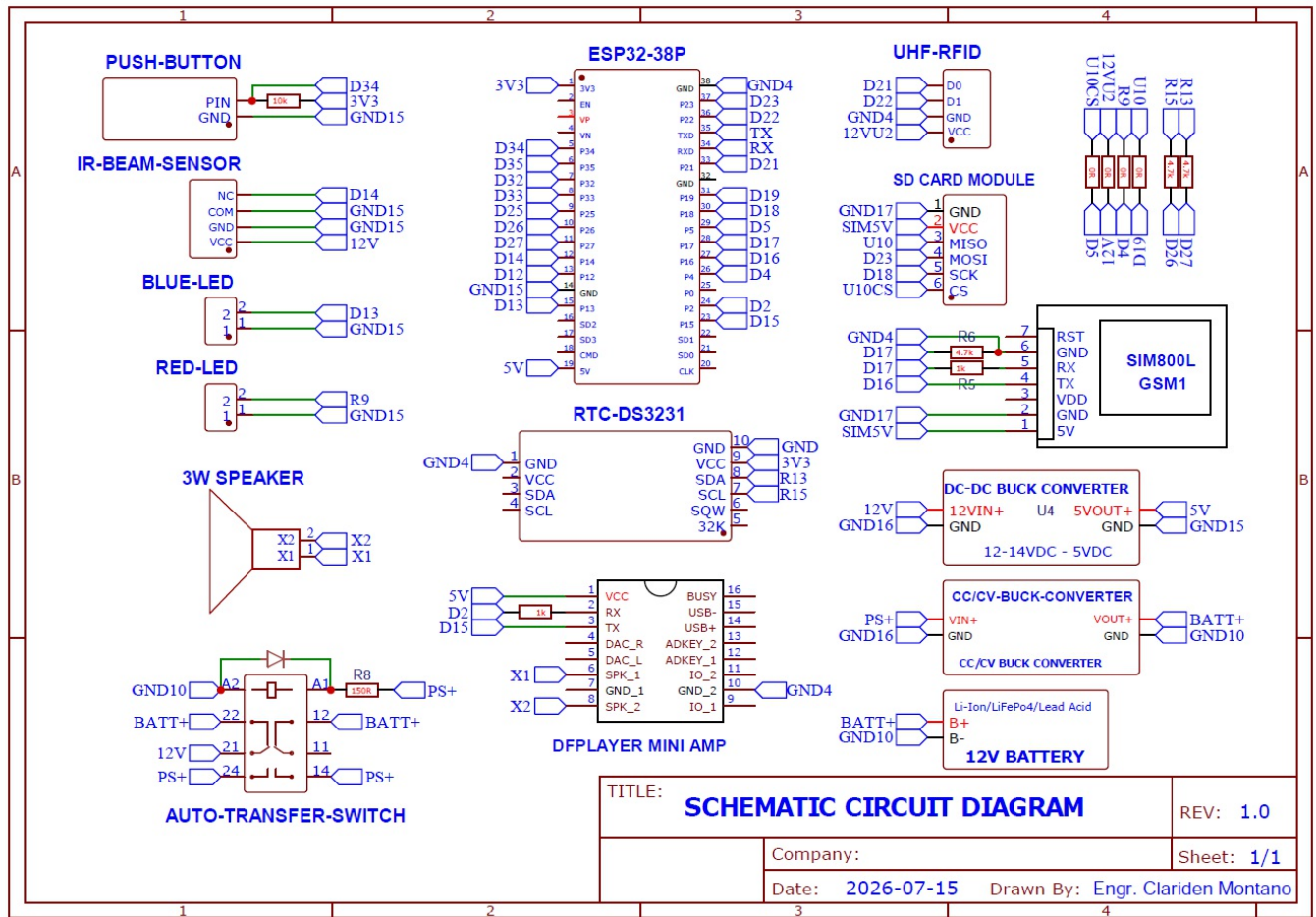
[!NOTE] Visual Reference Only

The realistic component layout below is for visual purposes and aesthetic representation. For exact pin mapping and electrical connections, strictly follow the official Schematic Circuit Diagram.



Schematic Circuit Diagram

For full hardware wiring instructions and component mapping, please refer to the official wiring schematic:



Optimized Pin Assignment (ESP32-38P)

All peripheral pairs are assigned to physically adjacent GPIO pins on the ESP32-38P to minimize wire crossings on a prototype board.

LEFT SIDE (top → bottom)	RIGHT SIDE
GPIO34 ← MODE BUTTON	GPIO23 ← SPI MOSI
GPIO35 ← (Unassigned)	GPIO22 ← WIEGAND D1
GPIO32 ← (Unassigned)	GPIO21 ← WIEGAND D0
GPIO33 ← (Unassigned)	GPIO19 ← SPI MISO
GPIO25 ← (Unassigned)	GPIO18 ← SPI SCK
GPIO26 ← RTC SCL	GPIO5 ← SD CS
GPIO27 ← RTC SDA	GPIO17 ← SIM800 TX
GPIO14 ← IR BEAM	GPIO16 ← SIM800 RX
GPIO12 ← (Unassigned)	GPIO4 ← Alarm LED (Red)
GND	GPIO0 ← (Unused, boot clock)
GPIO13 ← Status LED (Blue)	GPIO2 ← DFPlayer TX
	GPIO15 ← DFPlayer RX

2. How the System Works

The system has two main states or "modes". You can switch between them by holding down the **Mode Button** (GPIO 34).

2.1 Normal Mode (Guarding)

- **Blue Light:** Stays ON.
- **What it does:** The system uses the IR beam to watch the wall. If the beam is broken (meaning someone is jumping the wall), it triggers the alarm.
- **When triggered:** It plays a warning sound, flashes the red light, and sends a text message to a security guard.

2.2 Setup Mode

- **Blue Light:** Blinking fast.
- **How to enter:** Press and hold the Mode Button for 3 seconds.
- **What it does:** The system stops watching the wall and creates its own Wi-Fi network. You can connect to this Wi-Fi to change settings using your phone or laptop.
- **How to exit:** Press and hold the button for 3 seconds again. (It will also exit automatically after 1 hour if you forget).

2.3 Remote Control via Text Message (SMS)

You don't always have to press the physical button to change modes! You can send a text message to the system's SIM card to control it remotely:

- Text `AP ON`: Turns on the Wi-Fi Setup Mode so you can connect to the dashboard.
- Text `AP OFF`: Turns off the Wi-Fi Setup Mode and goes back to Normal Guarding mode.

(Note: For extra security, if "Number Lock" is enabled, the system will only obey text commands sent from the registered security guard's phone number!)

3. Changing Settings on the Dashboard

3.1 Connecting to the Setup Page

1. Put the system into **Setup Mode** (hold the button until the blue light blinks).
2. On your phone or laptop, connect to the Wi-Fi network:
 - **Wi-Fi Name:** AntiWallJump-Config
 - **Password:** configure123
3. A login page should pop up. If it doesn't, open a web browser and go to `http://192.168.4.1`.
4. The default admin password is admin.

3.2 What's on the Dashboard?

The setup page has different sections and buttons to help you control the system:

Top Menu Buttons (Quick Actions):

- **Sync Clock (Clock Icon):** Click this to instantly set the system's time to match your phone or computer.
- **Dark/Light Mode (Moon/Sun Icon):** Switch the screen between dark and light colors.
- **Logout (Door Icon):** Securely sign out of the setup page.
- **Stop Portal (Power Icon):** Quickly exit setup mode and turn the alarm back on right away.

Bottom Navigation Tabs:

- **Dashboard:** See the "Node Overview" (system health like the memory card status) and check the overall security status.
- **Registry:** Register student ID cards and see who triggered the alarm.
- **Settings:** Change the phone number for alerts, choose how close someone has to be to trigger the alarm, and pick which sound plays.
- **Records:** View a history of all past alarms and download a record file.

4. Student ID Cards (Registry)

The system keeps track of authorized students using their UHF RFID ID cards.

4.1 Adding a New Student Card

1. Go to the **Registry** tab on the dashboard.
2. Click **Scan Card**.
3. Tap a new ID card on the UHF RFID panel. The card's special ID will show up on the screen.
4. Type in the student's name and school ID number.
5. Click **Register** to save them in the system.

4.2 What happens when the alarm triggers?

- If the sensors spot someone climbing, it logs an **UNKNOWN INTRUDER**.
 - If someone holds a registered student ID card to the reader, the system grants them safe passage.
 - If an unregistered card is tapped, it triggers the alarm to keep the area secure.
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5. Alarm Records

The system saves a history of every time the alarm goes off. It saves this inside its internal memory and on a MicroSD card, so nothing gets lost!

5.1 Downloading the Records

You can download a spreadsheet file of all the past alarms:

1. Go to the **Records** tab on the dashboard.
2. Click **"Download Logs"**.
3. A file called `Campus_Security_Logs.csv` will download to your device. You can open this with Excel or Google Sheets to see a neat list of everything that happened.